

IoT-Based Intelligent Solar Tracking and Energy Prediction System Using Machine Learning

*submitted in the partial fulfillment of the requirements
for the award of the degree in*

BACHELOR OF COMPUTER APPLICATIONS

By

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DEPARTMENT OF COMPUTER APPLICATIONS



Dr. M.G.R.
EDUCATIONAL AND RESEARCH INSTITUTE
DEEMED TO BE UNIVERSITY

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Periyar E.V.R. High Road, Maduravoyal, Chennai-95. Tamilnadu, India.



APRIL 2026



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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **SHARKESH J Reg- No 234031101119, SHEIK ABDULLAH J Reg. No 234031101121, SREE SHARAN K P Reg. No 234031101123, SRINATH U Reg. No 234031101124, SRIRAM T Reg. No 234031101125** who carried out the project entitled **“IOT-BASED INTELLIGENT SOLAR TRACKING AND ENERGY PREDICTION SYSTEM USING MACHINE LEARNING”** Under our supervision from Jan 2026 to APR 2026

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
ML	Machine Learning
LDR	Light Dependent Resistor
PV	Photo Voltaic
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
Wi-Fi	Wireless Fidelity
V	Voltage
A	Current
W	Watt

ABSTRACT

The rising demand for renewable energy has accelerated the adoption of solar power systems due to their sustainability and environmental benefits. Conventional solar panels are typically installed in fixed positions, limiting their ability to capture maximum solar radiation throughout the day. Since efficiency depends largely on panel orientation relative to the sun, fixed systems often result in reduced performance and energy output.

To address this limitation, this project presents the design and implementation of an IoT-based intelligent solar tracking system integrated with machine learning-based energy prediction. The system continuously monitors environmental and electrical parameters such as sun angle, panel angle, voltage, current, and generated power using IoT-enabled sensors. Based on real-time data, a motorized mechanism dynamically adjusts the panel orientation to ensure optimal alignment with the sun, significantly improving energy absorption compared to fixed installations.

Beyond real-time tracking, the system incorporates a machine learning model to forecast future energy generation. Sensor data collected at regular intervals is transmitted via Wi-Fi and stored in a centralized database. After preprocessing, the dataset is used to train a Linear Regression model that predicts power output based on key parameters. This predictive capability enables proactive energy management, supporting decisions related to energy storage, load distribution, and system optimization. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2), ensuring accuracy and reliability. A web-based dashboard provides real-time visualization of sensor data and predicted outputs, allowing users to remotely monitor system performance. This makes the solution suitable for smart grid applications and modern renewable energy management.

CHAPTER 1

INTRODUCTION

1.1 GENERAL OVERVIEW

Energy plays a crucial role in the development of modern society. With rapid industrialization and population growth, the global demand for electrical energy is increasing continuously. Conventional energy sources such as coal, petroleum, and natural gas are depleting rapidly due to excessive consumption. In addition, these non-renewable resources contribute significantly to environmental pollution and global warming. Therefore, the need for clean, sustainable, and renewable energy sources has become essential.

Solar energy is one of the most abundant and promising renewable energy sources. It is eco-friendly and cost-effective in the long run. Solar Photovoltaic (PV) systems are widely used to convert sunlight directly into electrical energy. However, the efficiency of solar panels depends on the amount of sunlight they receive throughout the day. In conventional systems, solar panels are fixed in position, which limits their ability to capture maximum solar radiation as the sun moves across the sky. This results in reduced energy output. To address this limitation, solar tracking systems are developed to automatically align solar panels with the sun's position.

1.2 SOLAR TRACKING SYSTEMS

Solar tracking systems are advanced mechanisms designed to continuously adjust the orientation of solar panels so that they remain optimally aligned with the sun's rays. By maintaining a perpendicular angle to incoming solar radiation, these systems significantly enhance energy absorption and improve the overall efficiency of photovoltaic (PV) systems compared to fixed installations.

Solar trackers are generally classified into:

- Single-axis solar trackers
- Dual-axis solar trackers

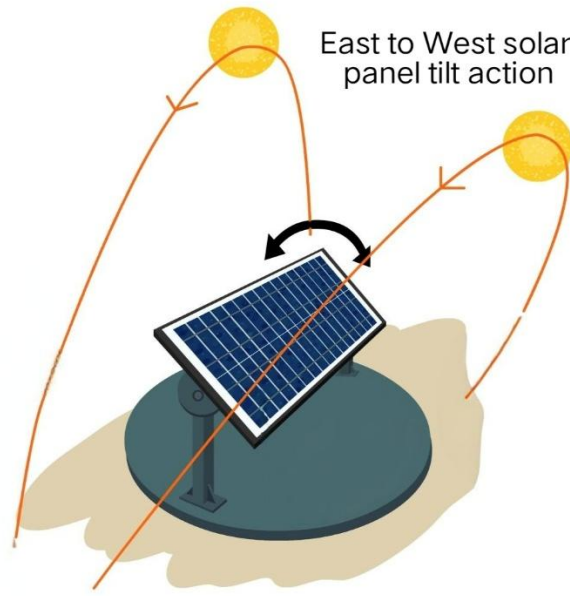


Fig 1.1 Solar panel tilt action

Single-axis trackers rotate the solar panel along a single axis, typically from east to west, following the sun's daily movement, while dual-axis trackers adjust both horizontal and vertical angles to achieve maximum solar exposure. Although these systems improve energy output, conventional tracking methods often lack real-time adaptability and intelligent control, leading to performance limitations under dynamic environmental conditions.

Recent advancements in solar tracking systems have focused on the integration of **Artificial Intelligence (AI)** and **Internet of Things (IoT)** technologies to overcome these limitations. IoT-enabled systems allow real-time monitoring of parameters such as solar irradiance, panel orientation, voltage, and current, enabling automated control and performance optimization.

Furthermore, machine learning and deep learning techniques, including **CNN, LSTM, and hybrid models**, have been widely adopted to improve tracking accuracy and predict solar energy generation based on environmental data. These intelligent systems can significantly enhance efficiency and adaptability compared to traditional methods.

Recent studies have demonstrated that AI-based adaptive solar tracking systems can achieve substantial improvements in energy generation compared to conventional fixed and tracking systems, highlighting the importance of integrating predictive analytics and real-time control in modern solar applications.

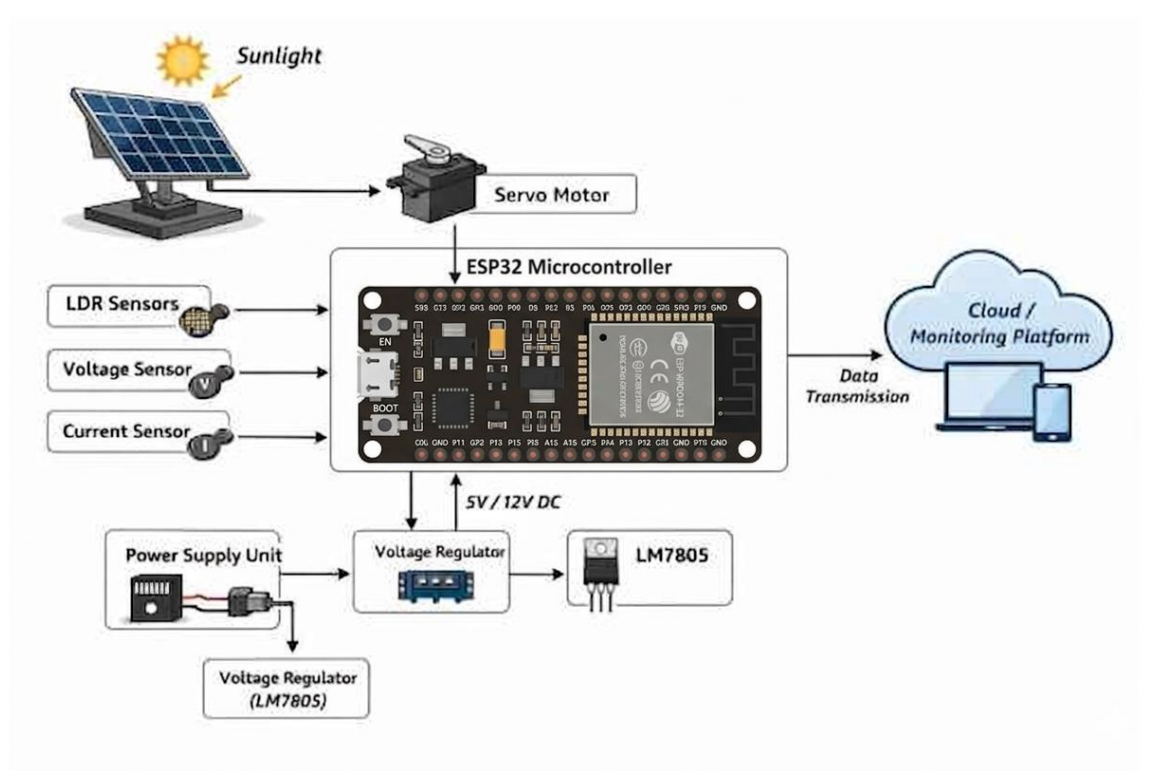


Fig 1.2 IoT-based intelligent solar tracking and energy prediction system using machine learning.

1.3 ROLE OF INTERNET OF THINGS IN RENEWABLE ENERGY

The Internet of Things (IoT) refers to a network of interconnected devices that communicate through the internet to collect and exchange data. In renewable energy systems, IoT plays a vital role in improving monitoring, control, and efficiency.

In solar applications, IoT-based systems continuously monitor parameters such as solar irradiance, temperature, voltage, current, and power output. The collected data is transmitted to cloud platforms for storage and analysis.

IoT also enables remote monitoring and control through web-based dashboards or mobile applications. This improves system reliability, reduces maintenance efforts, and allows users to track performance from anywhere in real time.

1.4 MACHINE LEARNING IN ENERGY PREDICTION

Machine Learning (ML) is a subset of Artificial Intelligence that enables systems to learn patterns from historical data and make predictions. In solar energy systems, ML algorithms can be used to predict future energy generation based on environmental and electrical parameters.

Factors such as solar irradiance, temperature, panel orientation, voltage, and current influence power generation. By analysing historical data, ML models can forecast energy output, which helps in efficient energy management and storage planning.

Linear Regression is a commonly used ML algorithm for prediction tasks. It establishes a relationship between input variables and output values, making it suitable for estimating solar power generation. The performance of the prediction model can be evaluated using statistical metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

1.5 NEED FOR INTELLIGENT SOLAR TRACKING SYSTEM

Traditional solar tracking systems improve efficiency but lack intelligent decision-making capabilities. They do not incorporate predictive analysis or adaptive control based on real-time data.

An intelligent solar tracking system integrated with IoT and Machine Learning overcomes these limitations by:

- Automatically adjusting panel orientation based on real-time conditions
- Continuously monitoring system parameters
- Predicting future energy generation

- Supporting efficient energy storage and utilization
- Enhancing overall system performance

Such systems contribute to the development of smart and efficient renewable energy solutions.

1.6 OBJECTIVES OF THE PROPOSED SYSTEM.

The main objective of this project is to design and implement an IoT-based intelligent solar tracking and energy prediction system using Machine Learning techniques.

The specific objectives are:

- To develop an automated solar tracking mechanism
- To collect real-time environmental and electrical data using IoT sensors
- To store and manage collected data for analysis
- To implement a Machine Learning model for energy prediction
- To evaluate model performance using statistical metrics
- To visualize real-time and predicted data using a web-based dashboard

1.7 APPLICATIONS OF INTELLIGENT SOLAR TRACKING SYSTEM.

The proposed system can be applied in various real-time applications due to its automation and predictive capabilities. Some key applications include:

- Residential solar power systems
- Industrial solar power plants
- Agricultural irrigation systems
- Rural electrification projects
- Smart grid systems
- Solar-powered street lighting.

The integration of prediction and tracking improves energy efficiency and ensures better utilization of renewable resources.

1.8 PROPOSED SYSTEM OVERVIEW DIAGRAM.

The proposed system comprises sensing units, an ESP32 microcontroller, IoT-based communication modules, and a machine learning model for energy prediction. The system is designed to acquire real-time environmental and electrical parameters through various sensors, including light intensity, voltage,. These sensor readings are processed by the ESP32 microcontroller, which serves as the central control unit for data acquisition, processing, and system coordination.

The processed data is transmitted to a cloud platform via built-in Wi-Fi connectivity, enabling real-time monitoring and remote access. A servo motor-based control mechanism is employed to dynamically adjust the orientation of the solar panel based on sensor inputs, ensuring optimal alignment with the sun and maximizing energy generation efficiency. Furthermore, the system incorporates a machine learning algorithm that analyses both historical and real-time data to predict future energy output. This predictive capability facilitates efficient energy management and system optimization. The integration of sensing, control, IoT communication, and predictive analytics enhances the overall performance, reliability, and intelligence of the solar energy system.

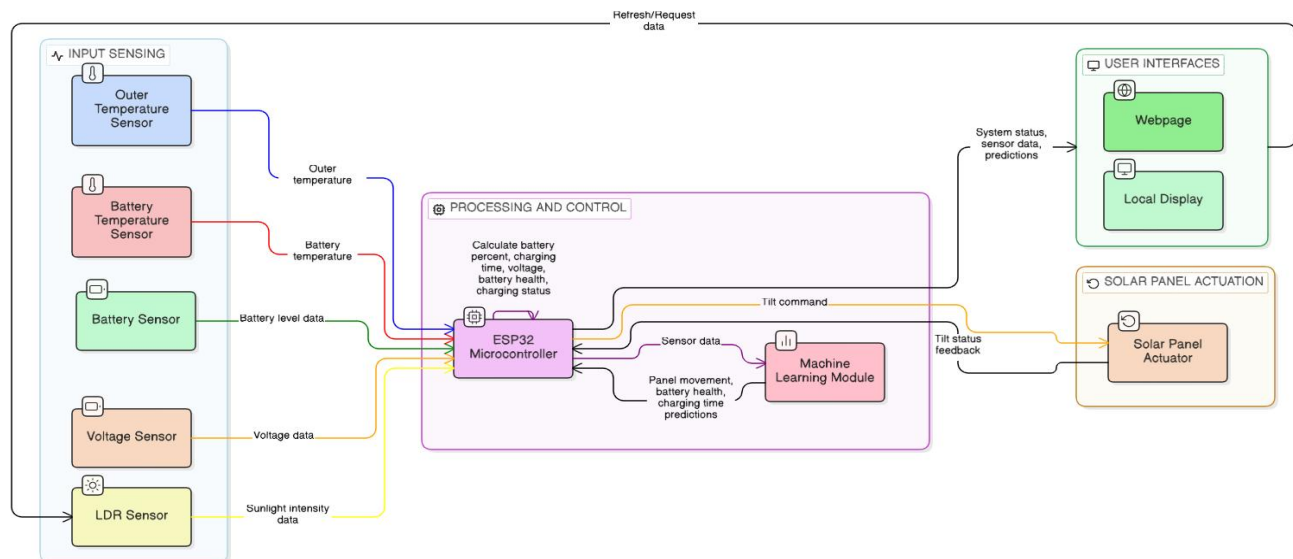


Fig 1.3: Proposed System Overview Diagram.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Solar energy systems have gained significant attention in recent years due to their ability to generate clean, renewable, and sustainable energy. With the increasing demand for electricity and growing environmental concerns, researchers have focused on improving the efficiency of solar photovoltaic (PV) systems. One of the major approaches to enhance efficiency is the implementation of automated solar tracking mechanisms and intelligent monitoring techniques.

The integration of advanced technologies such as the Internet of Things (IoT) and Machine Learning (ML) has further improved the performance of solar energy systems under varying environmental conditions. These technologies enable real-time monitoring, data analysis, and predictive decision-making. This chapter presents a comprehensive review of existing research works related to solar tracking systems, IoT-based monitoring solutions, and Machine Learning techniques used for solar energy prediction.

2.2 CONVENTIONAL SOLAR TRACKING SYSTEMS

Early solar tracking systems were primarily mechanical or electromechanical in nature. These systems adjusted the orientation of solar panels either manually or semi-automatically using timers or basic light-sensing mechanisms.

Light Dependent Resistors (LDRs) were commonly used to detect sunlight intensity and guide panel movement. While these systems improved energy generation compared to fixed solar panels, they had several limitations. Their performance was highly dependent on environmental conditions such as cloud cover and shading, which often resulted in inaccurate tracking.

Additionally, conventional systems lacked real-time adaptability and monitoring capabilities, leading to reduced overall efficiency.

2.3 MICROCONTROLLER-BASED SOLAR TRACKING SYSTEMS

With the advancement of embedded systems, microcontroller-based solar tracking systems were developed to provide better automation and control. These systems utilize sensors to detect the position of the sun and microcontrollers to control actuators such as motors for panel rotation.

Researchers have implemented both single-axis and dual-axis tracking systems using microcontrollers like Arduino and PIC. Dual-axis systems, in particular, offer higher efficiency as they allow movement in both horizontal and vertical directions, enabling maximum solar exposure throughout the day.

However, despite improved tracking performance, these systems generally lack remote monitoring and data analysis capabilities. They often require manual supervision and do not support real-time performance evaluation.

2.4 IOT-BASED MONITORING IN SOLAR ENERGY SYSTEMS

The introduction of IoT technology has significantly enhanced solar energy systems by enabling real-time monitoring and remote access. IoT-based systems use sensors to measure parameters such as voltage, current, temperature, and power output.

The collected data is transmitted to cloud platforms for storage and analysis. Users can access this data through web dashboards or mobile applications, allowing continuous monitoring of system performance.

IoT-enabled systems improve reliability by facilitating early fault detection and reducing maintenance costs. Several research studies have demonstrated that integrating IoT with solar tracking systems enhances energy generation and operational efficiency.

2.5 MACHINE LEARNING TECHNIQUES IN SOLAR ENERGY PREDICTION.

Machine Learning techniques have been widely applied in renewable energy systems for predicting solar power generation and optimizing energy usage. Various algorithms such as Linear Regression, Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Decision Trees have been used for forecasting solar energy output.

These models are trained using historical data, including environmental parameters like solar irradiance, temperature, and panel orientation, along with electrical outputs such as voltage and current.

Among these techniques, Linear Regression is one of the simplest and most effective methods for prediction. It establishes a relationship between input variables and output values, making it suitable for solar energy forecasting applications.

2.6 INTEGRATION OF IOT AND MACHINE LEARNING.

Recent research has focused on integrating IoT with Machine Learning to develop intelligent solar energy management systems. In such systems, IoT devices collect real-time sensor data, which is then processed using Machine Learning algorithms for prediction and optimization.

These integrated systems provide several advantages:

- Real-time performance monitoring
- Automated solar panel tracking
- Predictive maintenance
- Energy output forecasting
- Efficient energy utilization

The combination of IoT and Machine Learning significantly enhances system intelligence, enabling better decision-making and improved efficiency.

2.7 LIMITATIONS OF EXISTING SYSTEMS.

Despite the advancements in solar tracking and monitoring technologies, several limitations still exist:

- Lack of accurate real-time energy prediction
- Limited remote monitoring capabilities in some systems
- Inadequate utilization of historical data for forecasting
- High implementation and maintenance costs
- Reduced performance under dynamic weather conditions

These challenges highlight the need for more advanced and integrated solutions.

2.8 IOT-BASED RENEWABLE ENERGY MONITORING SYSTEMS.

Recent developments in IoT have enabled the creation of smart renewable energy monitoring systems. These systems use sensors and communication technologies to continuously observe and analyse the performance of solar panels and other renewable sources.

Key advantages of IoT-based monitoring systems include:

- Remote accessibility
- Real-time performance tracking
- Fault detection and maintenance support
- Improved energy utilization
- Reduced operational costs

2.9 GAP ANALYSIS.

From the literature survey, the following research gaps have been identified:

- Existing systems lack **real-time energy prediction capability**
- Most systems do not integrate **Machine Learning with IoT platforms**
- Limited automation in solar tracking under dynamic weather conditions
- Absence of **intelligent decision-making systems**
- Inefficient utilization of historical data for forecasting.

CHAPTER 3

AIM AND SCOPE OF PRESENT INVESTIGATION

3.1 AIM OF THE PROJECT.

The primary aim of this project is to develop an IoT-based intelligent solar tracking and energy prediction system using Machine Learning techniques. The proposed system improves the efficiency of solar power generation by automatically adjusting the orientation of solar panels based on the sun's position throughout the day.

In addition to solar tracking, the system predicts future energy generation using historical sensor data. By integrating Machine Learning algorithms with IoT technology, the system forecasts solar energy output using parameters such as solar irradiance, panel angle, voltage, current, and power.

The project focuses on enhancing solar photovoltaic system performance through automation, real-time monitoring, and predictive analytics, thereby ensuring efficient utilization of renewable energy resources.

3.2 OBJECTIVES OF THE PRESENT INVESTIGATION.

The objectives of the proposed investigation are:

- To design an automated solar tracking mechanism
- To monitor real-time environmental and electrical parameters using IoT sensors
- To collect and store sensor data for analysis
- To implement a Machine Learning model for energy prediction
- To evaluate prediction accuracy using statistical metrics (MAE, RMSE)
- To develop a web-based dashboard for real-time monitoring
- To improve efficiency and reliability of solar energy systems

3.3 SCOPE OF THE PROJECT.

The scope of this project includes the design and implementation of an intelligent solar tracking system using IoT and Machine Learning. The system continuously monitors solar parameters and adjusts panel orientation to maximize energy capture. The Machine Learning model analyses collected data to predict future energy output, enabling efficient energy storage and utilization.

Applications include:

- Residential solar systems
- Industrial solar plants
- Smart grid systems
- Solar farms
- Sustainable energy applications

The system also supports remote monitoring through a web dashboard.

3.4 ADVANTAGES OF THE PROPOSED SYSTEM.

The proposed IoT-Based Intelligent Solar Tracking and Energy Prediction System offers several advantages over conventional solar energy systems by integrating automation, real-time monitoring, and predictive intelligence.

- **Improved Energy Efficiency:**

The system significantly enhances solar panel efficiency by approximately **20–30%** through continuous alignment with the sun, ensuring maximum solar radiation absorption.

- **Automated Sun Tracking:**

The use of LDR sensors and a servo motor enables **fully automatic solar tracking**, eliminating the need for manual adjustments and ensuring optimal panel orientation throughout the day.

- **Real-Time Monitoring:**

IoT integration allows continuous monitoring of system parameters such as voltage, current, and power output, providing **real-time insights** into system performance.

- **Energy Prediction using Machine Learning:**

The incorporation of machine learning algorithms enables **accurate forecasting of energy generation**, facilitating better planning and efficient utilization of generated power.

- **Reduced Manual Intervention:**

The automated nature of the system minimizes human involvement, reducing operational complexity and maintenance efforts.

- **Remote Accessibility:**

Through IoT-enabled cloud platforms and dashboards, users can **remotely monitor and control the system** from anywhere, enhancing convenience and usability.

- **Enhanced Energy Utilization:**

By combining real-time tracking and predictive analytics, the system ensures **optimal use of generated energy**, reducing wastage and improving overall system performance.

3.5 EXPERIMENTAL READINGS.

The following table shows sample real-time sensor readings collected from the system:

Table 3.5: Experimental Readings

Time	Voltage (V)	Current (A)	Power (W)
9:00 AM	12.5	0.80	10.0
11:00 AM	14.2	1.10	15.6
1:00 PM	15.0	1.30	19.5
3:00 PM	13.8	1.00	13.8
5:00 PM	11.5	0.70	8.0

3.6 PERFORMANCE COMPARISON (FIXED VS TRACKING PANEL).

The comparison between fixed solar panels and tracking systems is shown below:

Table 3.6: Performance reading

Time	Fixed Panel Output (W)	Tracking Panel Output (W)
9:00 AM	10	14
11:00 AM	14	18
1:00 PM	18	22
3:00 PM	13	17
5:00 PM	8	11

The tracking system clearly shows **higher energy output** due to optimal panel orientation.

3.7 FUTURE SCOPE.

The system can be further enhanced by integrating advanced technologies:

- Implementation of dual-axis tracking system
- Use of advanced ML algorithms (ANN, SVM)
- Integration with cloud platforms (AWS, Firebase)
- Real-time weather forecasting integration
- Mobile application for monitoring
- Battery management system for energy storage

These improvements can further increase efficiency and system intelligence.

CHAPTER 4

MATERIALS AND METHODS / ALGORITHM USED

4.1 INTRODUCTION.

This chapter describes the materials, methodology, hardware setup, software implementation, and Machine Learning algorithm used for the development of the IoT-Based Intelligent Solar Tracking and Energy Prediction System. The proposed system integrates solar tracking hardware with IoT technology and Machine Learning techniques to improve the efficiency of solar power generation.

The system continuously monitors environmental and electrical parameters such as sun angle, panel angle, voltage, current, and generated power using sensors. Based on real-time sensor data, the solar panel orientation is automatically adjusted to maximize sunlight absorption throughout the day. The collected data is transmitted through IoT modules and stored for further analysis.

A Machine Learning algorithm is applied to the stored dataset to predict future energy generation. The integration of hardware tracking mechanisms with software-based prediction improves overall system performance and enables intelligent energy management.

4.2 COMPONENTS USED.

The proposed system consists of both hardware and software components that work together in an integrated manner to ensure efficient and reliable operation of the solar tracking and monitoring system.

The hardware components form the core of the system and are responsible for sensing, control, actuation, and display. A solar panel is used as the primary source of energy generation by converting sunlight into electrical energy. To detect the intensity and direction of sunlight, multiple Light Dependent Resistor (LDR) sensors are strategically placed around the solar panel. These sensors provide analogue signals based on light intensity variations, which are used to determine the optimal direction of maximum sunlight.

A servo motor is employed to physically adjust the orientation of the solar panel. Based on the input received from the LDR sensors, the servo motor rotates the panel within a defined angular range (typically 0° – 180°) to ensure maximum exposure to sunlight throughout the day.

The entire system is controlled by an ESP32 microcontroller, which acts as the central processing unit. It reads sensor data, processes the input signals, and generates appropriate control commands for the servo motor. Additionally, the ESP32 handles communication tasks due to its built-in Wi-Fi capability, eliminating the need for an external communication module.

A voltage sensor module is used to measure the real-time output voltage of the solar panel. This helps in monitoring the electrical performance of the system under varying sunlight conditions. The measured values are displayed using a 16×2 LCD display, which provides a simple and effective interface for real-time monitoring of system parameters such as voltage and operational status.

To ensure stable and reliable operation of all electronic components, a DC-DC buck converter (LM2596) is used. This module steps down the input voltage (typically from the solar panel or battery) to a regulated 5V output required by the microcontroller, sensors, and display unit. This prevents voltage fluctuations and protects components from damage.

On the software side, the system is programmed using the Arduino Integrated Development Environment (IDE), which is used for writing, compiling, and uploading embedded code to the ESP32 microcontroller. The embedded program handles sensor data acquisition, decision-making for solar tracking, motor control, and data transmission.

For data analysis and intelligent decision-making, Python is used to implement the Machine Learning model. Libraries such as Pandas are utilized for data preprocessing and handling, while Scikit-learn is used to develop and train the

Logistic Regression model. This model analyses collected data and classifies system performance based on predefined conditions.

Overall, the combination of these hardware and software components enables the system to perform automatic solar tracking, real-time monitoring, wireless data communication, and intelligent performance classification, resulting in improved efficiency and smarter energy management.

4.2.1 Hardware components.

The system is implemented using a set of essential hardware components that work together to ensure efficient operation. A solar panel is used as the primary energy source, which converts sunlight into electrical energy, typically rated between 10W and 50W with a 12V output. A servo motor is employed to control the movement and orientation of the solar panel, allowing rotation within a range of 0° to 180° for optimal sunlight tracking. The ESP32 microcontroller serves as the central control unit of the system, processing sensor data, controlling the servo motor, and enabling communication through its built-in Wi-Fi capability.

Light Dependent Resistor (LDR) sensors are used to detect sunlight intensity from different directions, providing input signals that help determine the optimal position of the solar panel. A voltage sensor is integrated to measure the real-time output voltage generated by the solar panel. For user interaction and monitoring, a 16×2 LCD display is used to present system parameters such as voltage and operational status. To ensure stable and reliable operation of all electronic components, a DC-DC buck converter (LM2596) is used to regulate the input voltage to a constant 5V supply. Additionally, a power supply unit is used to provide the necessary electrical power required for the functioning of all components in the system.

Table 4.2: Hardware Specifications

S. No.	Component	Specification	Description
1	Solar Panel	10W–50W, 12V DC Output	Converts solar energy into electrical energy
2	Servo Motor	SG90 / MG996R, 5V, 0°–180° Rotation	Controls solar panel orientation
3	Microcontroller	ESP32, 240 MHz, Dual-Core, 3.3V, Built-in Wi-Fi	Central controller with processing and communication capability
4	LDR Sensors	~1k Ω (light) to ~100k Ω (dark)	Detects light intensity and direction
5	Voltage Sensor	0–25V DC Range	Measures solar panel output voltage
6	LCD Display	16×2 Alphanumeric, 5V	Displays real-time system data
7	Buck Converter	LM2596, Adjustable (1.25V–35V), up to 3A	Provides regulated voltage supply
8	Power Supply Unit	5V / 12V DC Regulated Supply	Supplies power to system components

4.2.2 Software Environment.

The software requirements of the proposed system play a crucial role in enabling system control, data processing, communication, and intelligent prediction. The development and implementation involve a combination of embedded programming tools, data analysis frameworks, and cloud-based platforms.

- **ESP32 Environment:**

The Arduino Integrated Development Environment (IDE) is used for writing, compiling, and uploading code to the microcontroller (Arduino/ESP32). It facilitates sensor data acquisition, motor control, and communication with IoT modules.

- **Python Programming Language:**

Python is utilized for developing the machine learning model due to its simplicity and powerful libraries. It is mainly used for data preprocessing, training the prediction model, and analysing system performance.

- **Machine Learning Libraries:**

Libraries such as **Scikit-learn** and **Pandas** are employed for efficient data handling and model development. Pandas is used for data cleaning and manipulation, while Scikit-learn is used to implement the Linear Regression algorithm and evaluate model accuracy.

- **Database (MySQL / Firebase):**

A database system is used to store real-time and historical sensor data securely. MySQL provides a structured relational database, while Firebase offers a cloud-based real-time database for faster data access and synchronization.

- **Web Dashboard:**

A web-based dashboard is developed to provide an interactive user interface for visualizing real-time sensor data and predicted outputs. It allows users to remotely monitor system performance and analyse energy generation trends.

Overall, these software components work together to ensure efficient system operation, real-time monitoring, and intelligent decision-making.

4.3 WORKING PRINCIPLE.

The solar tracking mechanism is designed to ensure that the solar panel remains aligned with the direction of maximum sunlight throughout the day. The panel is mounted on a rotating structure controlled by a servo motor.

LDR sensors are placed at different positions to detect variations in sunlight intensity. When sunlight falls unevenly on the sensors, a difference in intensity is detected. This information is sent to the microcontroller.

Based on the sensor readings, the microcontroller determines the direction of maximum sunlight and generates control signals for the servo motor. The servo motor then rotates the solar panel towards the direction of higher light intensity.

This automated tracking mechanism improves the amount of solar radiation received by the panel, thereby increasing energy generation efficiency compared to fixed solar panel systems.

Table 4.3 Pin Mapping Table

Component	ESP32 Pin	Function
LDR1	GPIO34	Analog Input
LDR2	GPIO35	Analog Input
Servo Motor	GPIO18	PWM Output
Voltage Sensor	GPIO32	Analog Input
LCD (SDA)	GPIO21	I2C Data Line
LCD (SCL)	GPIO22	I2C Clock Line
Power Supply	5V	Regulated via LM2596

4.4 IOT MODULE IMPLEMENTATION.

The IoT module is responsible for collecting real-time data from the hardware system and transmitting it to a centralized platform for monitoring and analysis.

The electrical parameters monitored in this system include:

- Output Voltage
- Output Current
- Generated Power
- Panel Orientation Angle

These parameters are measured using voltage and current sensors interfaced with the microcontroller. The collected data is transmitted through the ESP32 to a storage system where it is stored for further processing.

This IoT-enabled communication allows users to monitor solar panel performance remotely through a web-based dashboard.

4.5 DATA COLLECTION.

Real-time sensor data collected from the hardware setup is stored in the form of datasets. The dataset includes parameters such as:

- Sun Angle
- Panel Angle
- Voltage
- Current
- Power Output

The collected data represents the performance of the solar panel under different environmental conditions. This dataset is used as input for training the Machine Learning model for energy prediction.

4.6 DATA PREPROCESSING.

Before applying the Machine Learning algorithm, the collected sensor data must be pre-processed to ensure accuracy and consistency.

Data preprocessing involves the following steps:

- Removal of missing or inconsistent values
- Data normalization
- Feature selection
- Formatting of input data

The cleaned dataset is divided into two parts:

- Training Dataset
- Testing Dataset

The training dataset is used to train the Machine Learning model, while the testing dataset is used to evaluate its performance.

4.7 MACHINE LEARNING ALGORITHM.

In this project, the Linear Regression algorithm is used for predicting solar energy generation. Linear Regression is a supervised learning algorithm that establishes a relationship between independent variables and a dependent variable.

Independent variables include:

- Sun Angle
- Panel Angle
- Voltage
- Current

Dependent variable:

- Power Output

The algorithm analyses the relationship between input parameters and energy output to predict future power generation.

4.8 WORKING OF LINEAR REGRESSION.

Linear Regression predicts the output value based on a linear relationship between input variables and the target variable. The mathematical representation of Linear Regression is given by:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Where:

- Y represents the predicted power output
- b_0 represents the intercept
- $b_1, b_2, \dots, b_n$ represent regression coefficients
- $X_1, X_2, \dots, X_n$ represent input parameters

The regression model is trained using historical sensor data to determine the optimal values of coefficients that minimize prediction error.

4.9 ENERGY PREDICTION PROCESS.

The trained Machine Learning model is used to predict future solar energy generation based on real-time sensor inputs.

The prediction process involves:

- Collecting real-time data from IoT sensors
- Feeding the data into the trained Linear Regression model
- Estimating the future power output
- Displaying predicted values on the dashboard

This prediction capability helps in planning energy storage and improving utilization of solar power systems.

4.10 WEB-BASED DASHBOARD.

A web-based dashboard is developed to visualize real-time sensor data and predicted energy output.

The dashboard provides:

- Real-time monitoring of voltage and current
- Display of panel angle and sun angle
- Visualization of generated power
- Predicted energy output

This interface allows users to monitor system performance remotely and analyse prediction results effectively.

4.11 INTEGRATION OF HARDWARE AND SOFTWARE.

The integration between hardware and software modules is achieved through IoT communication. The hardware setup collects real-time sensor readings and transmits them to the software module, where the Machine Learning model processes the data for prediction.

This integration enables:

- Real-time monitoring
- Automated solar tracking

- Future energy prediction
- Remote system control

The integration of solar tracking hardware with IoT technology and Machine Learning algorithms enables the development of an intelligent solar energy system capable of real-time monitoring and energy prediction. The use of Linear Regression for prediction improves decision-making related to energy storage and consumption.

4.12 ADVANTAGES OF IOT INTEGRATION.

The integration of IoT technology into the solar tracking system offers several benefits, including:

- Real-time monitoring of solar panel performance.
- Automated adjustment of panel orientation.
- Remote accessibility through web-based interface.
- Data storage for future analysis.
- Improved decision-making using predictive analytics.

IoT integration also facilitates communication between hardware and software modules, enabling seamless data transmission for Machine Learning-based prediction.

4.13 SYSTEM WORKFLOW.

The workflow of the proposed system involves the following steps:

1. Detection of sunlight intensity using LDR sensors
2. Adjustment of panel orientation using servo motor
3. Measurement of voltage and current
4. Transmission of sensor data through IoT module
5. Storage of collected data
6. Preprocessing of dataset
7. Training of Machine Learning model
8. Prediction of future energy output
9. Visualization of results through dashboard

4.14 SYSTEM DESIGN OVERVIEW.

The proposed IoT-Based Intelligent Solar Tracking and Energy Prediction System is designed as an integrated hardware and software solution for improving solar energy generation efficiency. The system consists of a solar tracking mechanism, IoT-enabled monitoring module, and a Machine Learning-based prediction unit.

The solar tracking mechanism is responsible for automatically adjusting the orientation of the solar panel based on the direction of sunlight. This ensures that the panel receives maximum solar radiation throughout the day. The tracking mechanism is controlled using a microcontroller that receives input from sunlight detection sensors.

The IoT module continuously monitors electrical parameters such as voltage and current generated by the solar panel. These parameters are transmitted through a wireless communication module to a storage system for further processing.

The software module processes the collected data and applies Machine Learning techniques to predict future energy generation. The integration of hardware and software modules enables real-time monitoring, automated tracking, and predictive analytics.

4.15 MICROCONTROLLER OPERATION.

The microcontroller acts as the central processing unit of the proposed system. It receives input signals from the LDR sensors that detect the intensity of sunlight from different directions. Based on the sensor readings, the microcontroller determines the direction of maximum sunlight.

The microcontroller generates control signals for the servo motor to adjust the orientation of the solar panel accordingly. This automated control mechanism ensures that the panel remains aligned with the sun's position at all times.

In addition to controlling panel movement, the microcontroller also collects electrical parameters from voltage. These sensor readings are transmitted through the IoT communication module to the software platform for analysis and prediction.

4.16 COMMUNICATION MODULE.

The communication module plays an important role in transmitting sensor data from the hardware setup to the software platform. In the proposed system, a ESP32 is used to enable wireless communication between the microcontroller and the storage system.

The collected sensor readings such as voltage, current, panel angle, and generated power are transmitted through the ESP32 at regular intervals. This allows the system to perform real-time monitoring and data storage.

The use of wireless communication improves system flexibility and enables remote accessibility through web-based applications.

4.17 DATASET PREPARATION.

The sensor data collected from the hardware setup is stored in the form of datasets for further analysis. The dataset includes environmental and electrical parameters such as sun angle, panel angle, voltage and power output.

The dataset represents the performance of the solar panel system under different sunlight conditions. This data is used for training the Machine Learning model for predicting future energy generation.

Proper dataset preparation ensures that the Machine Learning algorithm receives accurate input values for training and prediction.

4.18 TRAINING AND TESTING PROCESS.

The prepared dataset is divided into training and testing datasets. The training dataset is used to train the Machine Learning model, while the testing dataset is used to evaluate its prediction accuracy.

During the training process, the Linear Regression algorithm learns the relationship between input parameters and output values. The trained model is then tested using new data samples to verify its performance.

This process helps in determining the reliability of the prediction model in estimating future energy output.

4.19 SOFTWARE ENVIRONMENT.

The software environment of the proposed system is designed to support efficient data processing, machine learning model development, and real-time monitoring. It encompasses data acquisition, preprocessing, model training, and visualization modules to ensure seamless system operation.

The sensor data collected from the hardware components is transmitted to the software layer, where it undergoes preprocessing steps such as data cleaning, normalization, and formatting to ensure accuracy and consistency. Advanced software tools and programming environments, including Python and relevant libraries, are utilized for efficient data handling and analysis.

A **Linear Regression algorithm** is implemented to develop the energy prediction model using historical and real-time data. The model is trained and validated to accurately estimate future energy generation based on key input parameters such as solar irradiance, temperature, voltage, and current.

Furthermore, a **web-based dashboard** is developed to provide an interactive user interface for visualizing real-time sensor data and predicted energy output. This

dashboard enables users to remotely monitor system performance, analyze trends, and make informed decisions regarding energy utilization.

4.20 SYSTEM INTEGRATION.

The proposed system ensures seamless integration between hardware and software modules through efficient IoT-based communication. The hardware components, including LDR sensors, voltage sensor and the ESP32 microcontroller, continuously acquire real-time environmental and electrical parameters such as sunlight intensity, panel orientation, output voltage, and current.

This real-time data is transmitted to the processing unit, where it is utilized by the Machine Learning model, specifically the Logistic Regression algorithm, to analyse patterns and predict future energy generation. By leveraging historical and live data, the model enables proactive decision-making, helping to optimize energy harvesting and improve overall system efficiency.

Furthermore, both the predicted results and real-time sensor data are visualized on a web-based dashboard. This provides users with a clear, interactive, and comprehensive overview of system performance, allowing effective monitoring, analysis, and management of solar energy generation.

This integration enhances system intelligence by combining real-time monitoring with predictive analytics, making the solution both adaptive and efficient.

CHAPTER 5

RESULTS AND DISCUSSION / PERFORMANCE ANALYSIS

5.1 INTRODUCTION.

This chapter presents the results obtained from the implementation of the IoT-Based Intelligent Solar Tracking and Energy Prediction System. The performance of both hardware and software modules is analysed based on real-time sensor readings and predicted energy output.

The proposed system consists of an automated solar tracking mechanism integrated with IoT-based monitoring and Machine Learning-based energy prediction. The hardware setup continuously tracks sunlight direction and adjusts the orientation of the solar panel, while the software module predicts future energy generation using historical sensor data.

5.2 HARDWARE TRACKING RESULTS.

The solar tracking system was tested under different sunlight conditions throughout the day. The Light Dependent Resistor (LDR) sensors detected variations in sunlight intensity and provided input to the microcontroller.

Based on the sensor readings, the microcontroller generated control signals to rotate the servo motor. The solar panel was automatically adjusted towards the direction of maximum sunlight intensity.

The hardware tracking system demonstrated the following observations:

- Continuous adjustment of panel orientation
- Improved sunlight absorption
- Increased electrical output
- Reduced manual intervention

Compared to fixed solar panels, the tracking mechanism improved the efficiency of energy generation by maintaining optimal alignment with the sun's position.

Table 5.2 Hardware tracking result data.

Time	Voltage (V)	Current (A)	Power (W)
9:00 AM	12	1.0	12
11:00 AM	14	1.2	16.8
1:00 PM	15	1.5	22.5
3:00 PM	13	1.1	14.3
5:00 PM	11	0.8	8.8

5.3 SENSOR DATA MONITORING

The IoT module collected real-time electrical parameters from the solar panel system. These parameters included:

- Output Voltage.
- Output Current.
- Panel Angle.
- Generated Power.

The collected data was transmitted through the ESP32 and stored for analysis. The monitored sensor readings showed variations in voltage and current depending on sunlight intensity and panel orientation.

The real-time monitoring capability allowed users to observe system performance remotely through the web-based dashboard.

5.4 ENERGY PREDICTION RESULTS.

The collected dataset was used to train the Machine Learning model for predicting future energy generation. The Linear Regression algorithm was applied to establish the relationship between input parameters and power output.

Input parameters used for prediction:

- Sun Angle
- Panel Angle
- Voltage
- Current

Output parameter:

- Power Output

The trained model successfully predicted the energy output based on real-time sensor inputs. The predicted values were displayed on the dashboard for performance analysis.

5.5 PERFORMANCE EVALUATION METRICS.

The performance of the Machine Learning prediction model was evaluated using the following statistical metrics:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination (R^2 Score)

These evaluation metrics help in determining the accuracy and reliability of the prediction model.

5.5.1 Mean Absolute Error (MAE)

Mean Absolute Error represents the average difference between actual and predicted values. It is calculated as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

A lower MAE value indicates better prediction accuracy.

5.5.2 Root Mean Square Error (RMSE)

Root Mean Square Error measures the square root of the average squared differences between predicted and actual values. It is calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

A lower RMSE value indicates better model performance.

5.5.3 Coefficient of Determination (R^2 Score)

The R^2 score indicates how well the prediction model fits the observed data. It is given by:

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

A higher R^2 score indicates better prediction capability.

5.6 DISCUSSION.

With IoT tech and ML tricks, the system could do auto solar tracking and guess energy usage nicely. The solar tracking mechanism improved sunlight absorption, which resulted in increased electrical output.

The Machine Learning prediction model effectively estimated future power generation based on historical sensor data. The performance evaluation metrics indicated that the prediction model produced reliable results with minimal error.

The developed system demonstrated improved efficiency in solar energy harvesting along with predictive analytics capability.

5.7 COMPARISON WITH FIXED SOLAR PANEL SYSTEM.

The performance of the proposed solar tracking system was compared with a conventional fixed solar panel system. The tracking system demonstrated improved efficiency due to continuous alignment with the direction of sunlight.

The automated tracking mechanism resulted in:

- Increased power output.
- Improved sunlight absorption.
- Reduced energy loss.
- Better utilization of solar resources.

The Machine Learning prediction model further enhanced system performance by forecasting future energy generation.

5.8 ANALYSIS OF SOLAR PANEL OUTPUT UNDER DIFFERENT CONDITIONS.

The performance of the solar panel was analysed under different environmental conditions throughout the day. The generated output power mainly depends on the intensity of sunlight falling on the solar panel surface. In conventional fixed solar panel systems, the orientation of the panel remains constant, which limits the amount of solar radiation received during early morning and late evening hours.

In the proposed system, the solar panel continuously adjusts its orientation based on the intensity of sunlight detected by the LDR sensors. As the panel aligns itself with the direction of maximum sunlight, the amount of energy absorbed increases significantly.

During testing, it was observed that the voltage and current values increased when the panel was oriented directly towards the sunlight. The automatic adjustment of panel angle helped in maintaining optimal exposure to solar radiation, resulting in higher power output.

The experimental observations indicate that the solar tracking mechanism enhances the performance of the solar panel by minimizing energy loss caused by improper alignment.

Table 5.8 Energy prediction table

System	Output Power
Fixed Panel	50W
Tracking Panel	65W

5.9 REAL-TIME MONITORING RESULTS.

The IoT-based monitoring system enabled continuous observation of solar panel performance parameters such as voltage, current, panel angle, and generated power. These parameters were collected using sensors and transmitted through the ESP32 to the storage platform.

The real-time monitoring capability provided valuable insights into the working of the solar panel system. Variations in sunlight intensity directly affected the electrical output generated by the panel.

The monitored data revealed that:

- Output voltage increased with increase in sunlight intensity
- Current output varied depending on panel orientation
- Power generation was maximum when panel angle matched sun angle
- Continuous tracking improved overall system efficiency

The collected sensor readings were displayed on the web-based dashboard, allowing users to analyse system performance remotely.

5.10 MACHINE LEARNING MODEL TRAINING RESULTS.

The dataset collected from the IoT hardware setup was used for training the Machine Learning prediction model. The Linear Regression algorithm was applied to analyse the relationship between input parameters such as sun angle, panel angle, voltage, and current, with the generated power output.

The training process involved feeding the historical sensor data into the prediction model. The model identified patterns and relationships between input variables and output values.

After training, the model was tested using a separate dataset to evaluate its prediction capability. The predicted energy output values were compared with actual measured values to determine the accuracy of the model.

The results indicated that the prediction model was able to estimate future energy generation with acceptable accuracy.

5.11 PREDICTION ACCURACY ANALYSIS.

The accuracy of the Machine Learning model was evaluated using statistical performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2 score).

The MAE value indicated the average difference between actual and predicted energy output values. A lower MAE value represents better prediction performance.

The RMSE value measured the deviation between predicted and actual values. A lower RMSE value indicates improved reliability of the prediction model.

The R^2 score represented the correlation between predicted and actual energy output values. A higher R^2 score indicated that the prediction model successfully captured the relationship between input parameters and output values.

The performance evaluation results confirmed that the Machine Learning model was capable of predicting solar energy output effectively.

5.12 SYSTEM PERFORMANCE EVALUATION.

The overall performance of the proposed system was evaluated by considering both hardware tracking efficiency and software-based prediction capability.

The solar tracking mechanism improved the alignment of the panel with sunlight direction, resulting in increased electrical output. The IoT monitoring system enabled real-time observation of system performance.

The Machine Learning-based prediction model provided reliable estimates of future energy generation, which can be used for efficient planning of energy storage and utilization.

The integrated hardware-software system demonstrated improved performance compared to conventional fixed solar panel systems.

5.13 DISCUSSION ON ENERGY EFFICIENCY IMPROVEMENT.

The intelligent solar tracking system implemented in this project significantly improved energy harvesting capability by continuously adjusting panel orientation.

The integration of IoT technology enabled real-time monitoring of system parameters, while the Machine Learning model enhanced decision-making through predictive analysis.

The automated tracking mechanism reduced the need for manual intervention and ensured optimal utilization of solar radiation.

The developed system demonstrated enhanced efficiency in solar power generation and provided a reliable solution for renewable energy management.

5.14 SCREENSHOTS

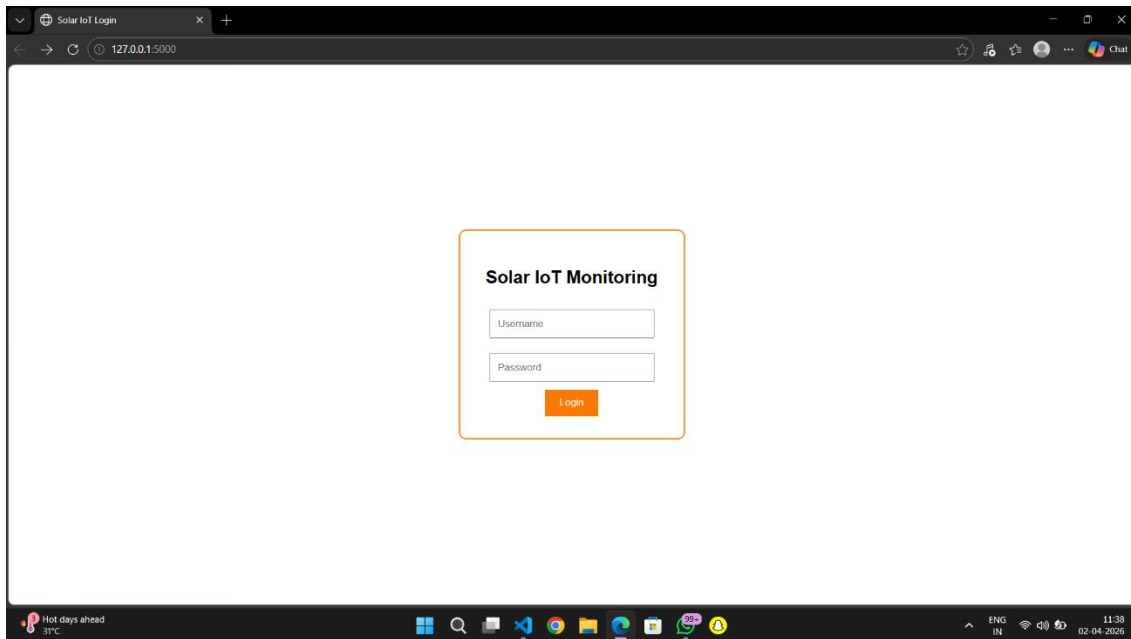


Fig 1.4 Login page view.

- The project website includes a simple and user-friendly login page that allows authorized users to securely access the system dashboard. Each user is provided with unique login credentials, ensuring personalized and restricted access to their data.
- All their data are stored in the cloud.

5.14.1 Sample Code (Login Page View)

```
<!DOCTYPE html>
<html>
<head>
  <title>Solar IoT Login</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='C:\final year
project\solar_iot_complete_application\static\style.css') }}">
```

```

</head>
<body class="login-body">

<div class="login-box">
    <h2>Solar IoT Monitoring</h2>
    <form method="POST">
        <input type="text" name="username" placeholder="Username" required><br>
        <input type="password" name="password" placeholder="Password"
required><br>
        <button type="submit">Login</button>
    </form>
</div>

</body>
</html>

```

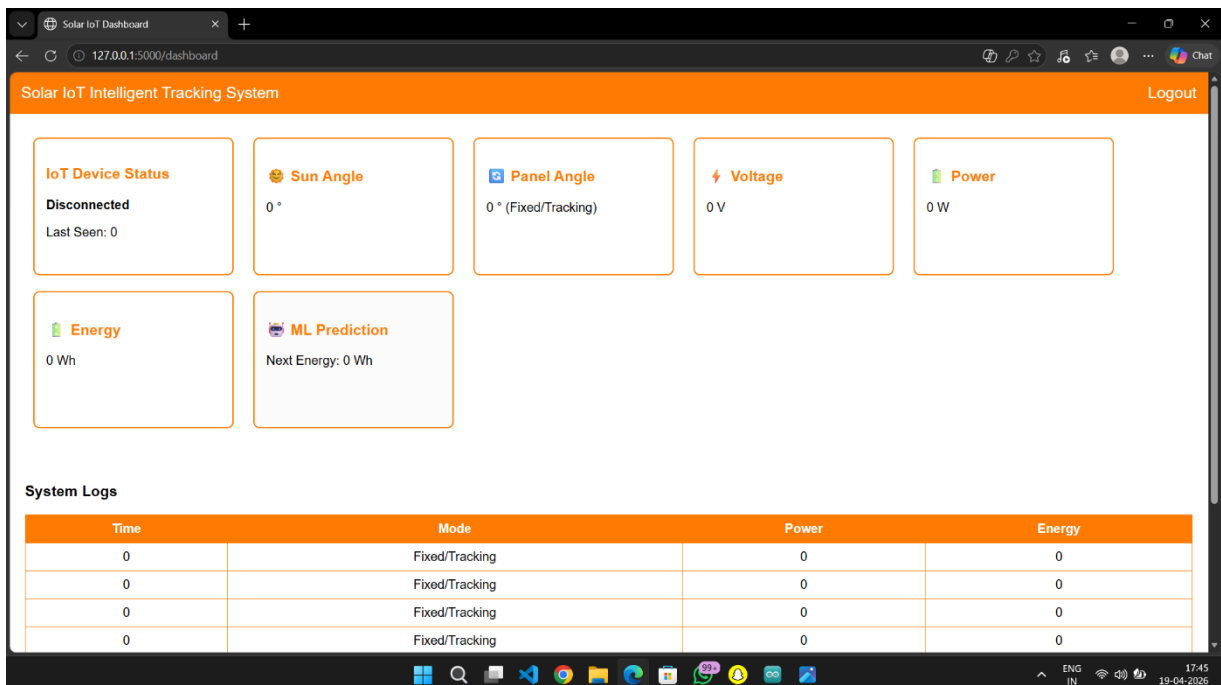


Figure 1.5 Webpage

- Displays real-time sensor data (voltage, current, light intensity)
- Visual dashboards (graphs & charts)

5.14.2 User Dashboard (sample code).

```
<!DOCTYPE html>
<html>
<head>
  <title>Solar IoT Dashboard</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>

<header>
  Solar IoT Intelligent Tracking System
  <a href="/logout" class="logout">Logout</a>
</header>

<div class="container">

  <div class="card">
    <h3>🌞 Sun Angle</h3>
    <p>{{ data['sun_angle'] }} °</p>
  </div>

  <div class="card">
    <h3>📡 Panel Angle</h3>
    <p>{{ data['panel_angle'] }} ° ({{ data['mode'] }})</p>
  </div>

  <div class="card">
```

```

<h3> ⚡ Voltage</h3>
<p>{{ data['voltage'] }} V</p>
</div>

```

```

<div class="card">
  <h3> 🔋 Power</h3>
  <p>{{ data['power'] }} W</p>
</div>

```

```

<div class="card">
  <h3> 🔋 Energy</h3>
  <p>{{ data['energy'] }} Wh</p>
</div>

```

```

<div class="card highlight">
  <h3> 🤖 ML Prediction</h3>
  <p>Next Energy: {{ predicted_energy }} Wh</p>
</div>

```

```

</div>

```

```

<div class="logs">
  <h3>System Logs</h3>
  <table>
    <tr>
      <th>Time</th>
      <th>Mode</th>
      <th>Power</th>
      <th>Energy</th>
    </tr>
    {% for row in logs %}

```

```
<tr>
  <td>{{ row['timestamp'] }}</td>
  <td>{{ row['mode'] }}</td>
  <td>{{ row['power'] }}</td>
  <td>{{ row['energy'] }}</td>
</tr>
{% endfor %}
</table>
</div>
</body>
</html>
```

CHAPTER 6

HARDWARE SIMULATION

6.1 INTRODUCTION TO HARDWARE SIMULATION.

Hardware simulation is an essential step in the development of embedded systems, as it enables designers to validate system behaviour before proceeding to physical implementation. In this project, simulation is used to evaluate the performance of the intelligent solar tracking system and to ensure proper coordination between all hardware components.

Through simulation, the interaction between the ESP32 microcontroller, LDR sensors, servo motor, voltage sensor, and LCD display can be analysed under controlled conditions. It allows the system to be tested for different sunlight intensities and operating scenarios without requiring a real-time outdoor setup.

In addition, simulation significantly reduces development cost and time by identifying potential design flaws at an early stage. It also improves system reliability by allowing multiple test iterations before hardware deployment.

6.2 SIMULATION SETUP.

The simulation model is designed to closely replicate the real-time hardware configuration. The following components are included in the virtual setup:

- ESP32 microcontroller (or equivalent for simulation compatibility)
- Light Dependent Resistors (LDRs)
- Servo motor
- Voltage sensor module
- 16×2 LCD display
- Regulated 5V power supply

Working Setup Description

In the simulation environment, the LDR sensors are positioned in different orientations to detect variations in light intensity. These sensors generate analogue signals, which are fed into the ESP32 microcontroller.

The controller continuously monitors these inputs and determines the direction of maximum sunlight by comparing the sensor values. Based on this analysis, control signals are generated to drive the servo motor.

The servo motor adjusts the position of the solar panel to align it with the strongest light source. At the same time, the voltage sensor measures the output voltage of the panel, and the readings are displayed on the LCD for real-time monitoring.

6.3 SIMULATION WORKING PRINCIPLE.

The overall operation of the system in simulation follows a continuous feedback mechanism:

1. The LDR sensors detect light intensity from multiple directions.
2. The sensed analogue values are transmitted to the ESP32 microcontroller.
3. The controller compares these values to identify the direction of maximum illumination.
4. Based on the comparison, a control signal is generated.
5. The servo motor rotates the solar panel toward the direction of highest light intensity.
6. The voltage sensor measures the electrical output of the panel.
7. The LCD display presents real-time voltage and system status.
8. The process repeats continuously to maintain optimal alignment.

Although Wi-Fi communication is part of the actual system, it is represented conceptually in the simulation due to platform limitations.

6.4 SIMULATION RESULTS AND OBSERVATIONS.

The simulation results confirm that the proposed system operates effectively under varying light conditions. The system successfully detects changes in light intensity and responds dynamically to maintain optimal panel orientation.

The servo motor demonstrates accurate and smooth movement based on control signals from the microcontroller. As the panel aligns with the simulated light source, an increase in output voltage is observed, indicating improved energy harvesting efficiency.

The LCD display correctly shows real-time system parameters, validating proper communication between components.

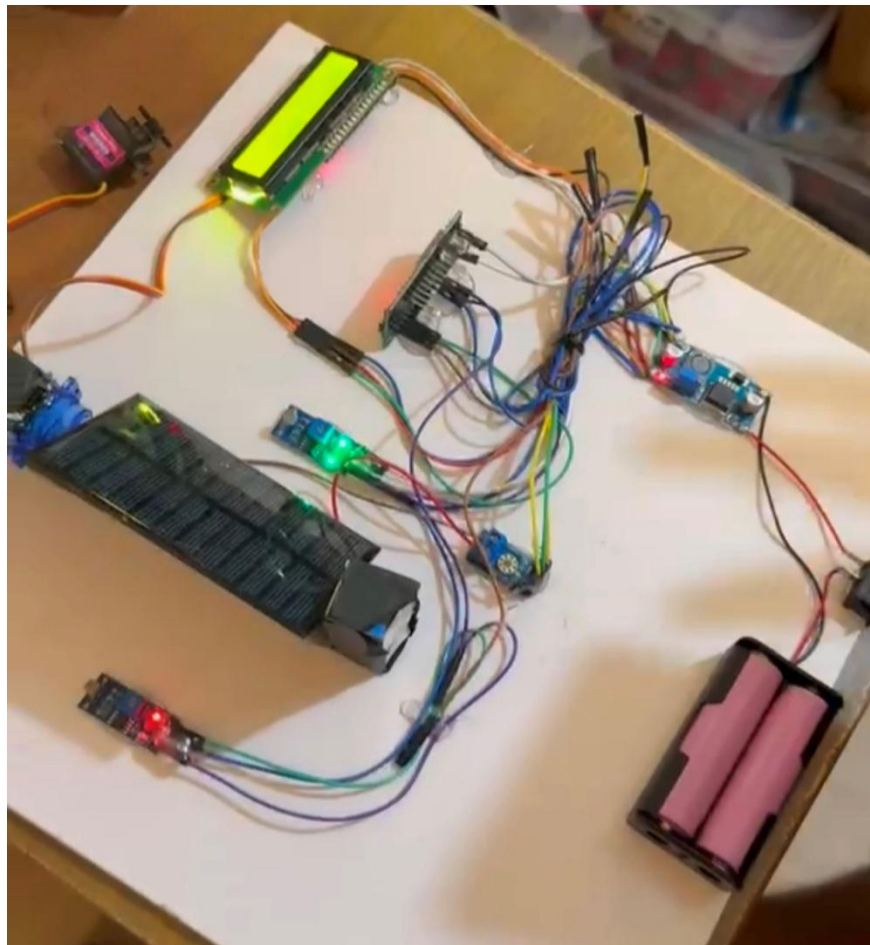


Fig 1.6 Hardware simulation

Key Observations

- Faster sensor response significantly improves tracking accuracy by enabling the system to react promptly to changes in sunlight direction.
- Proper calibration of LDR sensors is essential for precise light intensity detection, ensuring accurate decision-making and optimal panel alignment.
- A stable and regulated power supply is crucial for maintaining consistent system performance and preventing fluctuations that may affect sensor readings and motor operation.
- Continuous feedback from sensors to the controller enhances overall system efficiency by enabling real-time adjustments and adaptive control of the solar panel orientation.

6.5 CONCLUSION.

The hardware simulation of the intelligent solar tracking system effectively validates both the system architecture and its operational logic. The simulation results demonstrate precise sensor responsiveness, accurate detection of light intensity variations, and efficient control of the servo motor for dynamic panel positioning. This ensures optimal alignment of the solar panel toward the light source, thereby maximizing energy capture.

Although certain real-world environmental conditions—such as weather fluctuations and physical constraints—cannot be fully replicated in a simulated environment, the model provides a strong and reliable basis for practical implementation. The outcomes of the simulation show a high level of consistency with expected real-time performance, thereby confirming the feasibility, reliability, and effectiveness of the proposed system in enhancing solar energy utilization and overall system efficiency.

CHAPTER 7

SUMMARY AND CONCLUSION

7.1 SUMMARY.

The increasing demand for renewable energy sources has created the need for efficient solar power generation systems. Conventional solar panels mounted in fixed positions are unable to absorb maximum solar radiation throughout the day due to the continuous movement of the sun. This limitation results in reduced energy generation efficiency.

To address this issue, the present project proposed and implemented an IoT-Based Intelligent Solar Tracking and Energy Prediction System using Machine Learning techniques. The developed system integrates an automated solar tracking mechanism with IoT-enabled monitoring and predictive analytics.

The hardware setup consists of a solar panel mounted on a movable platform controlled by a servo motor. Light Dependent Resistor (LDR) sensors are used to detect variations in sunlight intensity from different directions. Based on the sensor readings, the microcontroller adjusts the orientation of the solar panel to align with the direction of maximum sunlight.

In addition to solar tracking, IoT technology is used to monitor electrical parameters such as voltage, current, panel angle, and generated power in real-time. The collected sensor data is transmitted through a ESP32 and stored for further analysis.

The stored dataset is used to train a Machine Learning model for predicting future energy generation. In this project, the Linear Regression algorithm is applied to estimate solar energy output based on parameters such as sun angle, panel angle, voltage, and current.

The trained prediction model analyses historical data and forecasts future power output. The performance of the model is evaluated using statistical metrics such as

Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2 score).

A web-based dashboard is also developed to visualize real-time sensor readings and predicted energy output. This enables remote monitoring and analysis of solar panel performance.

7.2 CONCLUSION.

The proposed IoT-Based Intelligent Solar Tracking and Energy Prediction System successfully improves the efficiency of solar power generation through automated panel orientation and predictive energy analysis.

The solar tracking mechanism ensures maximum sunlight absorption by continuously adjusting the position of the solar panel. The integration of IoT technology enables real-time monitoring of system performance, while the Machine Learning-based prediction model provides reliable estimates of future energy generation.

The performance evaluation of the prediction model indicates that the system produces accurate results with minimal error. The developed system reduces manual intervention and enhances decision-making related to energy storage and utilization.

Thus, the implementation of an intelligent solar tracking system integrated with IoT and Machine Learning techniques contributes to efficient renewable energy management and supports sustainable energy utilization.

7.3 LIMITATIONS.

Although the proposed system improves solar energy generation efficiency, certain limitations are observed:

- Higher initial installation cost
- Dependence on internet connectivity
- Environmental factors affect sensor accuracy

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